INTEGRAL UNIVERSITY
LUCKNOW (UTTAR PRADESH)

Power BI

Student Performance Analysis

Turning Data into Better Learning Outcomes

- Track Performance
- Find Insights
- Take Action



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Assignment : 2

Course :
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Better Insights... Brighter Future...

Project Overview

A data-driven view of academic performance, attendance and outcomes.

STUDENT RECORDS	AVERAGE MARKS	AVG. ATTENDANCE	PASS RATE
500	65.68	77.62	96%

01 Executive Summary

This project demonstrates how student-related data can be organized, queried and visualized to review academic performance, attendance, grade distribution and department-level patterns. SQL supports structured data analysis, while Power BI Desktop presents the results through KPI cards, charts and slicers.

02 Introduction

Student performance analysis helps summarize academic outcomes across a student group. A dashboard makes broad patterns easier to identify, while SQL provides a repeatable way to filter, aggregate and examine records. This project combines both approaches in an academic demonstration.

03 Project Objectives

- Organize student records in a structured dataset.
- Apply SQL concepts including table creation, filtering, aggregation, grouping and conditional analysis.
- Summarize marks, attendance, grades, departments and pass/fail outcomes.
- Build a Power BI dashboard with KPI cards, charts and slicers.
- Present findings clearly for academic review.

Data note: The project uses synthetic/sample data for learning and demonstration; it does not represent real institutional student records.

Dataset & SQL Analysis

Structured records make the analysis repeatable and easier to validate.

04 Dataset Description

The dataset contains 500 synthetic student records. It is intended for learning and demonstration only. Fields described in the project materials include student identifiers, department, gender, marks/average marks, attendance percentage, grade and result. Exact field names depend on the supplied CSV and SQL script.

Student ID	**Department**	**Gender**
Average Marks	**Attendance %**	**Grade / Result**

05 SQL Work

The SQL script is designed to create a database/table, load student records and run analysis queries. Typical outputs include total student count, average marks, average attendance, grade counts, department-wise averages, pass/fail counts and identification of students who may need academic support. The project defines a support indicator as attendance below 75% or average marks below 50.

COUNT STUDENT RECORDS

```
SELECT COUNT(*) AS total_students
FROM student_performance;
```

AVERAGE MARKS

```
SELECT AVG(Average_Marks) AS average_marks
FROM student_performance;
```

DEPARTMENT-WISE AVERAGE

```
SELECT Department, AVG(Average_Marks) AS avg_marks
FROM student_performance
GROUP BY Department;
```

GRADE DISTRIBUTION

```
SELECT Grade, COUNT(*) AS student_count
FROM student_performance
GROUP BY Grade;
```

Power BI Dashboard

Visual summaries help explore the dataset at a glance.

The supplied dashboard screenshot contains four KPI cards—Total Students, Average Marks, Average Attendance and Pass Percentage—along with a department-wise average marks bar chart, gender distribution doughnut chart, attendance-versus-marks scatter chart, and Grade / Department slicers.

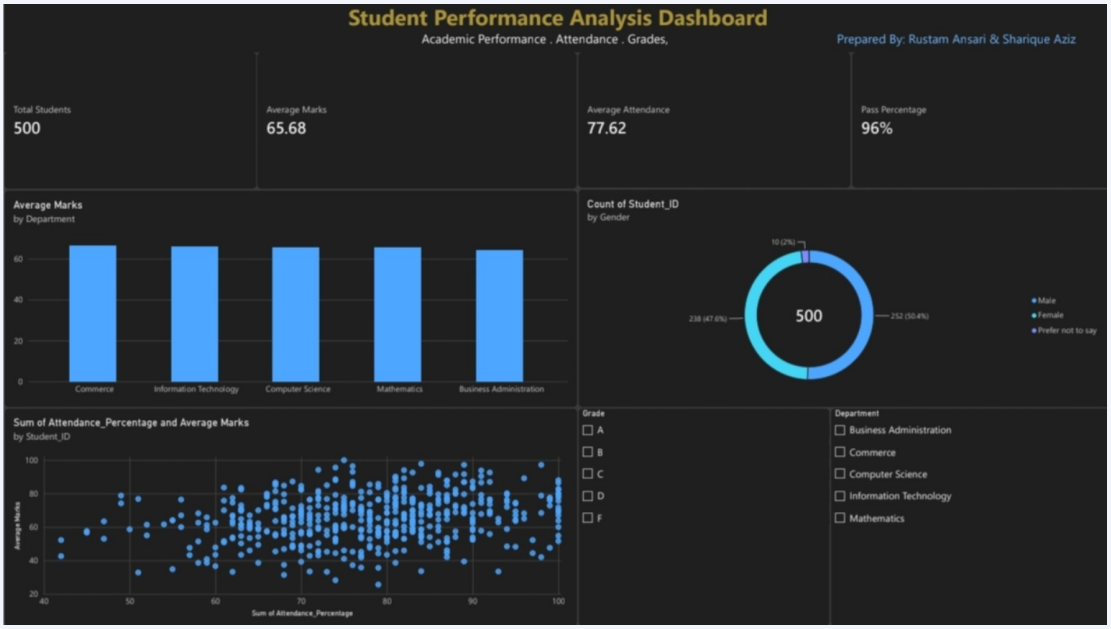


Figure 1. Student Performance Analysis dashboard supplied with the project.

06 Key Observations

500 student records represented

65.68 displayed overall average marks

77.62 displayed average attendance value

96% displayed pass percentage, rounded

- Department averages appear relatively close in the bar chart; exact values can be checked in Power BI.
- The gender chart includes Male, Female and Prefer not to say categories.
- The scatter chart can help inspect whether attendance and marks move together; visual association does not prove causation.

Project Methodology

How the Student Performance Analysis project moves from structured data to an interpretable dashboard.

Methodology at a glance

1

Dataset preparation

Use 500 synthetic student records for demonstration. The described fields include student ID, department, gender, average marks, attendance, grade and result.

2

Database and SQL analysis

Create the student-performance table and load records in the database environment. Use SQL to count records, calculate averages, group by department or grade, and summarize pass/fail outcomes.

3

Performance indicators

Review total students, average marks, average attendance and pass percentage. The project also defines a possible academic-support flag: attendance below 75% OR average marks below 50.

4

Power BI visualization

Present headline indicators as KPI cards, compare department average marks with a bar chart, show gender distribution with a doughnut chart, and explore attendance versus marks with a scatter chart.

5

Interpretation and validation

Use Grade and Department slicers to explore subsets. Check SQL outputs and visual aggregations before drawing conclusions; a visual association does not establish causation.

Why this methodology fits the objectives

The workflow links the project objectives to practical outputs: structured records support repeatable queries; SQL summarizes marks, attendance and outcomes; and Power BI makes the results easier to review through KPI cards, charts and slicers.

Data note: the source project describes the dataset as synthetic/sample data. These results should be treated as a learning demonstration, not as evidence about a real institution.

Project Outcomes & Recommendations

The outcomes below reproduce the headline values shown in the supplied project PDF and clarify what can—and cannot—be concluded from them.

Headline outcomes

500	65.68	77.62	96%
Student records	Displayed average marks	Displayed average attendance value	Displayed pass percentage (rounded)

What the dashboard helps show

• **Overall performance:** KPI cards provide a quick summary of record count, average marks, attendance and pass percentage.

• **Department comparison:** the bar chart supports comparison of average marks by department; the supplied report describes the department averages as relatively close, with exact values available in Power BI.

• **Category distribution:** the gender chart includes Male, Female and Prefer not to say categories, while the Grade slicer can be used to explore grade groups.

• **Attendance and marks:** the scatter chart can help inspect a possible relationship, but it does not prove that attendance causes higher or lower marks.

Recommendations before submission

• Run the SQL script in the required database environment and confirm that the returned totals and averages match the dashboard.

• Check the attendance field formatting. The source screenshot shows 77.62 but does not clearly establish whether the KPI should display a percent sign.

• For a student-level attendance-versus-marks scatter plot, confirm that each dot represents one student and review the attendance aggregation; the supplied screenshot labels the horizontal axis as a sum of attendance.

• If the project is extended, add term-wise comparisons, validated real-world data (with appropriate permission), and clearly tested academic-risk indicators.

Outcome statement

The project demonstrates an end-to-end analytics workflow: organize student records, query them with SQL, and communicate summary indicators through a Power BI dashboard. Its main outcome is a reusable demonstration of how academic data can be summarized and explored; the figures should be validated in the original database and dashboard before final submission.

Validation & Conclusion

Interpret results carefully and verify the dashboard before submission.

07 Limitations & Validation Notes

- This is a demonstration project based on synthetic data, not real institutional student data.
- The report summarizes the dashboard as shown in the supplied screenshot.
- The scatter chart horizontal axis is labelled as a sum of attendance. For a student-level comparison, consider an appropriate aggregation and confirm that each dot represents one student.
- Confirm whether the attendance KPI should be formatted with a percent sign; the supplied screenshot does not clearly establish this.
- Verify SQL execution and outputs in the database environment required by the university before submission.

08 Conclusion

The Student Performance Analysis project demonstrates an end-to-end workflow: preparing structured data, using SQL to summarize records, and presenting key metrics through a Power BI dashboard. The dashboard offers a compact view of marks, attendance, pass rate, department averages and category distributions. Possible future enhancements include term-wise comparisons, validated real-world data and additional academic-risk indicators.



Project team
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THANK YOU

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